State of the Practice for Open-Source Neural Network Software

Department of Computing and Software
McMaster University
Hamilton, Ontario, Canada

Abstract

Open-source neural network (NN) software plays an important role in modern AI research and development. Assessing the software-engineering practices of NN packages, however, is challenging because relevant evidence is distributed across public artifacts such as documentation, tests, issue trackers, releases, and continuous integration (CI) configurations.

This paper presents DomainX, a tool-supported workflow for conducting a state-of-practice assessment of open-source NN software based on a published methodology. DomainX supports evidence capture, scoring through a measurement template spanning nine software-quality attributes, incorporation of repository-based measures, and overall comparison using the Analytic Hierarchy Process (AHP).

We apply this workflow to a curated shortlist of 28 open-source NN software packages and produce a traceable dataset containing template responses, repository-based measures, and package-level references (project website and repository URLs). Under the current configuration, TensorFlow achieved the highest overall score, followed by Catboost, OpenCV, PyTorch, and Hugging Face Transformers. Beyond the top-ranked package, many libraries were more tightly clustered, with the strongest differentiation arising from installability and correctness and verifiability. A companion repository provides the DomainX tool and associated assessment artifacts to support review and reuse.

Keywords: state of practice; research software; neural network libraries; software quality; measurement template; Analytic Hierarchy Process (AHP); traceability; reproducibility; repository mining.

1. Introduction

Research and engineering outcomes increasingly depend on software, and the quality of that software affects reliability, maintainability, and reproducibility. In domains where open-source libraries are widely reused, understanding current development practice is an important step toward improving it. A *state-of-practice* assessment provides a structured way to characterize what developers commonly do in a domain by evaluating representative software packages against explicit software-quality criteria using traceable evidence.

Conducting such assessments rigorously is challenging in practice. Evidence relevant to software quality is distributed across many public artifacts, including documentation, tutorials, repository history, tests, issue trackers, releases, and continuous integration (CI) configurations. This makes the assessment process time-consuming and creates challenges for auditability, maintenance, and reproducibility as repositories evolve over time.

This paper presents a tool-supported state-of-practice assessment for *neural network (NN) software*. The target audience is developers and maintainers of NN software; accordingly,

the scope is restricted to NN packages rather than general-purpose NN-supporting libraries from adjacent communities. The assessment workflow follows a published state-of-practice methodology based on (i) a measurement template organized by software-quality attributes, (ii) repository-based measures extracted from public repositories, and (iii) an Analytic Hierarchy Process (AHP) ranking procedure to combine evidence into transparent comparisons.

The contributions of this work are:

- A curated shortlist of 28 open-source NN software packages evaluated using a consistent, evidence-based procedure (Appendix Appendix A).
- A traceable assessment dataset recording measurement-template responses, repository-based measures, and package-level references (project website and repository URLs) for auditability and reuse.
- DomainX, a tool that supports evidence capture, scoring, ranking computations, and export of results to reduce manual effort and improve traceability (`https://github.com/thaafei/DomainX`).

The remainder of this paper defines key concepts and summarizes background (Section 2), describes the assessment methodology and tool support (Section 3), and reports results and observations from applying the approach to the shortlisted packages (Section 4). The paper concludes with discussion, recommendations, and limitations (Sections 5–7).

2. Background and Definitions

In this paper, *research software* refers to software used to support scientific or engineering work such as data processing, model training or inference, experimentation, and analysis. A *domain* is a coherent problem area characterized by shared goals and typical workflows; in a state-of-practice assessment, the domain determines which software packages are relevant and what evidence is meaningful for evaluation. A *library* (or package) is a reusable software component intended to be integrated into other programs. We use *evidence* to mean publicly available artifacts that indicate software-development practice, including documentation and tutorials, repository history (commits and releases), issue trackers and pull requests, tests, and continuous integration (CI) configurations.

The *state of the art* describes the most advanced methods and recommended best practices reported in research and practitioner literature. The *state of practice* describes what is commonly done in real projects: the tools used, the artifacts produced, and the quality-related practices that are routinely applied. Gaps between state of the art and state of practice are common, particularly in research settings where incentives and timelines may prioritize functionality and novelty over long-term maintainability, transparency, or reproducibility. A state-of-practice assessment aims to characterize this reality using explicit criteria and traceable evidence, enabling systematic comparison and targeted recommendations.

A published state-of-practice methodology guides our assessment by combining (i) a measurement template grounded in software-quality attributes with (ii) evidence extracted from public artifacts and repository-based measures, enabling comparisons that are intended to be reproducible and auditable Smith et al. (October 2021).

To produce an overall comparison across multiple criteria, we use the Analytic Hierarchy Process (AHP), a multi-criteria decision-making method that derives weights from pairwise comparisons of criteria and aggregates scores accordingly. AHP is well suited to this setting

because it makes weighting assumptions explicit and supports transparent trade-offs among software-quality attributes. In this work, AHP is used to compute an evidence-based ranking and to support sensitivity analysis on how rankings change under different weighting choices.

3. Methods

3.1. Domain framing and scope

Neural network (NN) software is defined here as open-source libraries whose primary purpose is to enable the development, training, deployment, analysis, or direct use of neural network models. The target audience for this assessment is developers and maintainers of NN software. Accordingly, general-purpose NN-supporting packages (e.g., numerical computing, data manipulation, or plotting libraries) are out of scope because they belong to different developer communities and would confound interpretation of NN software development practice. Packages are included when they have a publicly accessible repository and sufficient publicly available artifacts (e.g., documentation and repository history) to support evidence-based evaluation.

3.2. Candidate identification and filtering

Candidate packages were compiled by aggregating package names from authoritative sources and normalizing duplicates and naming variants. To align the assessment with the intended audience, packages that were NN-supporting rather than NN software were removed. Following supervisor guidance, NN software packages were retained even when they appeared only once across the authoritative sources to prioritize completeness of coverage. The final shortlist contains 28 packages. The full shortlist and excluded candidates (with brief reasons) are provided in Appendix Appendix A.

3.3. Evidence sources and data collection

Evidence was derived from publicly available artifacts associated with each package, including documentation (e.g., README and installation guides), tutorials/examples, repository history, releases, issue trackers and pull requests, tests, and continuous integration (CI) configuration when available. Repository-based measures were extracted using automated scripts/tools. Some measurement-template questions required manual judgment (e.g., installation and usage guidance, presence of tests, or CI configuration). Responses were recorded directly in the measurement template, and each package entry includes a project website URL and a source-code repository URL that can be used to verify the recorded responses against the corresponding public artifacts.

3.4. Measurement template and scoring

The assessment uses a published measurement template organized by software-quality attributes. The qualities assessed are: installability; correctness and verifiability; surface reliability; surface robustness; surface usability; maintainability; reusability; surface understandability; and visibility/transparency. For each package, publicly available evidence was used to answer template questions and derive per-quality scores according to the template's scoring rules. Scores were normalized and aggregated to support cross-package comparison. When evidence was missing or ambiguous, responses were recorded conservatively and flagged for traceability.

3.5. Repository-based measures

Repository-based measures complement the measurement template by capturing observable project characteristics and development activity from public repositories. Using the same repository-measure framework defined in the methodology paper, the collected raw data includes GitHub-derived measures such as stars, forks, watchers, open and closed pull requests, and open and closed issues, together with repository statistics such as number of developers, programming languages, file counts, lines of code, comment lines, blank lines, binary files, total commits, and commit activity over recent yearly and monthly intervals Smith et al. (October 2021). The methodology identifies `git_stats` and `scc` as the tools used to collect repository statistics, with GitHub repository data used when the project is hosted on GitHub Smith et al. (October 2021).

In addition to raw measures, the methodology defines a small set of processed measures derived from the raw data. These include project status (alive or dead), percentage of issues closed, and percentage of code that is comments Smith et al. (October 2021). Together, these measures provide additional context for interpreting the quality scores recorded in the measurement template.

3.6. Ranking method (AHP) and sensitivity analysis

To produce an overall comparison across multiple quality criteria, this work uses the Analytic Hierarchy Process (AHP), following the methodology paper Smith et al. (October 2021). AHP is a multi-criteria decision-making technique based on pairwise comparisons. In this context, it is used to compare software packages using the quality scores derived from the measurement template. The pairwise comparisons are used to construct matrices that yield an overall score for each package under the selected criteria weights Smith et al. (October 2021).

As emphasized in the methodology, the ranking is not meant to identify a single universally "best" package. Instead, it is intended to provide insight into the top-performing set of packages and to support interpretation of domain-wide patterns Smith et al. (October 2021). The methodology also includes sensitivity analysis so that the stability of the ranking can be examined under uncertainty or variation in the quality scores and weighting choices Smith et al. (October 2021).

3.7. Tool support: DomainX

DomainX supports a tool-based workflow for conducting the assessment and managing the resulting dataset. It provides an interactive interface for recording measurement-template responses and repository-based measures across the shortlisted packages. DomainX also integrates the Analytic Hierarchy Process (AHP) computation used to aggregate quality-level scores into an overall comparison and visualizes the resulting ranking. In addition, the tool supports export of assessment artifacts for reporting and review. The DomainX implementation is available at `https://github.com/thaafei/DomainX`.

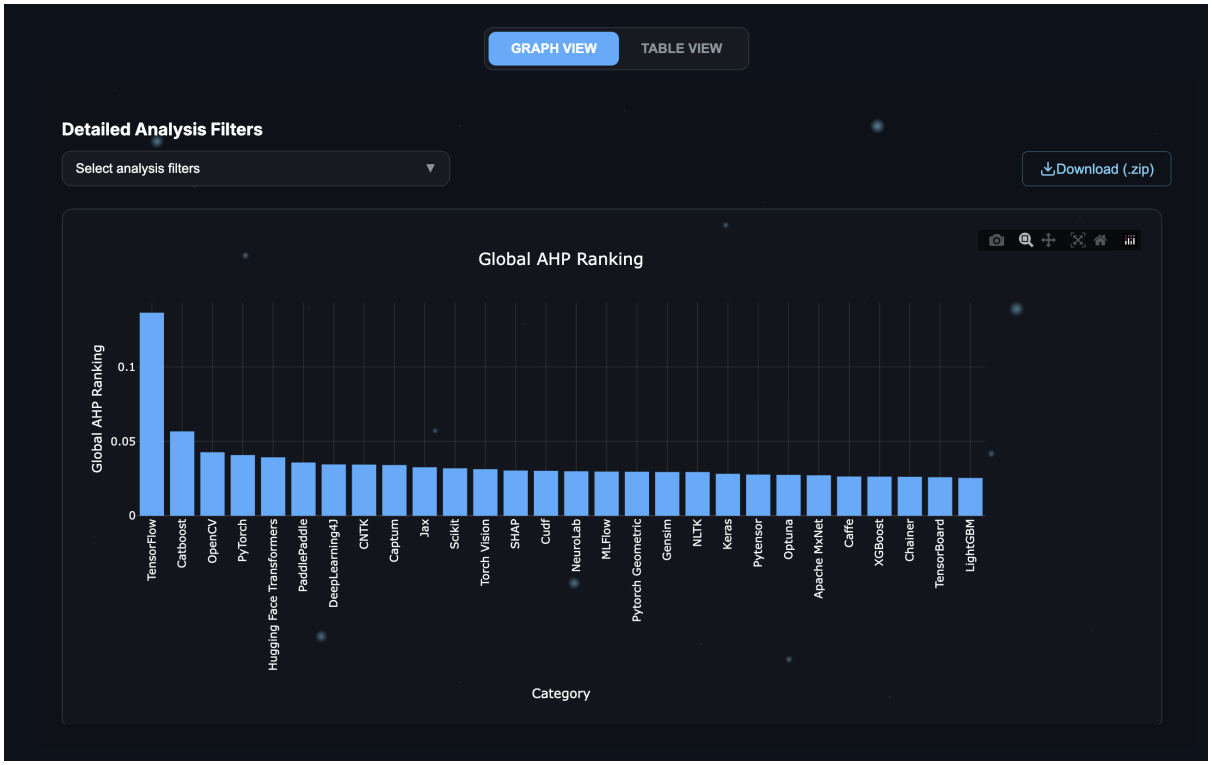

Figure 1: DomainX graph view illustrating the overall AHP ranking for the assessed packages.

3.8. Environment and reproducibility

Following the methodology, installation-based assessment is performed in a clean, isolated environment so that software packages are evaluated on a more level playing field Smith et al. (October 2021). The methodology recommends using virtual machines (VMs) because they provide a fresh environment for each installation attempt, reduce conflicts with pre-existing libraries, and help avoid "works-on-my-computer" effects Smith et al. (October 2021).

In the methodology, repository-based data is collected first, followed by installation and measurement-template scoring. The methodology also recommends recording environment details for each installation context, including the operating system and virtualization environment, to support repeatability Smith et al. (October 2021). In the present work, some older packages required additional environment setup because their dependencies and installation requirements were not readily compatible with a modern default environment. In these cases, containerized environments defined through Dockerfiles were used to reproduce a more suitable setup and improve consistency across installation attempts. These choices were made to improve consistency and transparency of the assessment process.

4. Results

4.1. Descriptive summary of assessed packages

The final shortlist contains 28 open-source neural network software packages selected according to the scope and filtering process described in Section 3. Based on the project-status information recorded in the assessment template, 23 packages were classified as alive and 5 were classified as dead. This mix of active and inactive projects provides important context for interpreting the assessment results, since project activity can affect the availability of documentation, testing artifacts, maintenance signals, and other observable evidence of development practice.

4.2. Overall ranking

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Figure 2 presents the overall AHP ranking for the shortlisted packages as produced by DomainX. TensorFlow achieved the highest overall score (13.65%), followed by Catboost (5.67%), OpenCV (4.27%), PyTorch (4.08%), and Hugging Face Transformers (3.93%). At the lower end of the ranking, LightGBM (2.54%), TensorBoard (2.59%), Chainer (2.62%), XGBoost (2.63%), and Caffe (2.64%) obtained the smallest overall scores.

A notable feature of the ranking is the large gap between TensorFlow and the rest of the shortlisted packages. Beyond the top-ranked package, many libraries are clustered within a comparatively narrow score range, suggesting that the middle and lower portions of the ranking are more densely packed. Under the current configuration, many category-level scores are identical across packages, so differences in the overall ranking are driven primarily by variation in installability, correctness and verifiability, and repository-based measures.

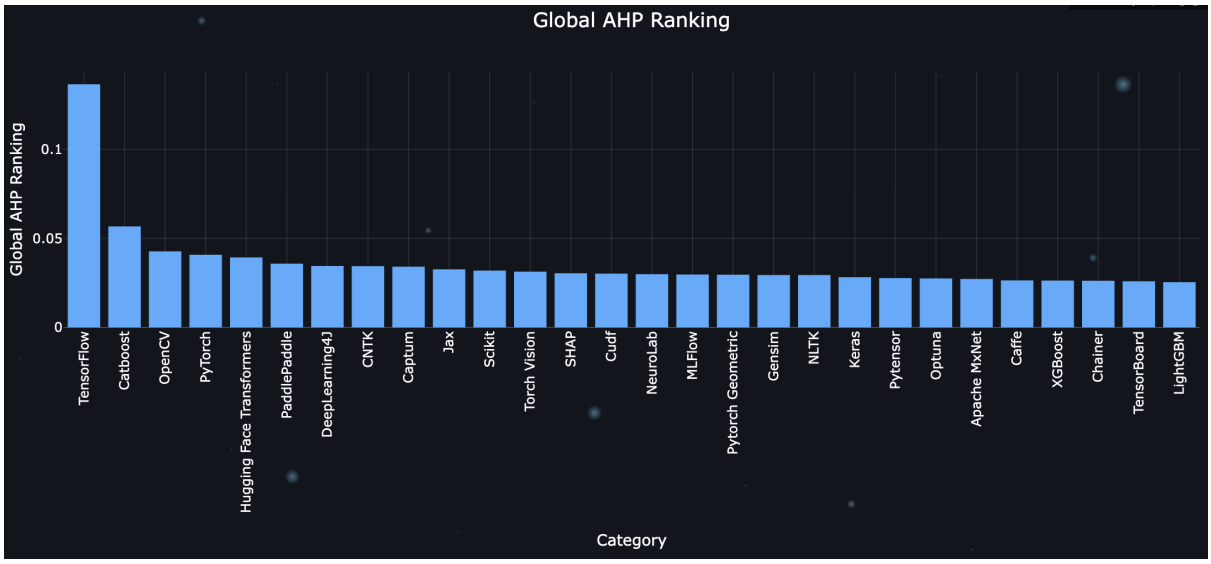

Figure 2: Overall AHP ranking for the shortlisted packages as visualized in DomainX.

Rank	Package	Overall Score
1	TensorFlow	13.65%
2	Catboost	5.67%
3	OpenCV	4.27%
4	PyTorch	4.08%
5	Hugging Face Transformers	3.93%
6	PaddlePaddle	3.58%
7	DeepLearning4J	3.45%
8	CNTK	3.44%
9	Captum	3.41%
10	Jax	3.26%

Table 1: Top 10 packages by overall AHP score.

Table 1 lists the top 10 packages under the current overall AHP ranking. TensorFlow is the clear top-ranked package, while the remaining top 10 packages are more tightly clustered.

4.5. Quality-by-quality breakdown

Figure 3 summarizes the per-quality results across the 28 shortlisted packages. The heatmap shows that several assessed categories take nearly the same value across most of the shortlisted packages. In particular, surface reliability, surface robustness, surface usability, reusability, visibility/transparency, and surface understandability are highly uniform in the current results. As a result, the most visible differences across packages arise in installability, quality, and correctness and verifiability.

Correctness and verifiability has the strongest visible effect on the ranking. TensorFlow obtains the highest value in this category (99.98%), substantially exceeding the other shortlisted packages, and this contributes strongly to its position as the top-ranked package overall.

Installability also differentiates several packages. Catboost records the highest installability value (18.18%), followed by Captum (11.36%) and a group of packages including DeepLearning4J, NeuroLab, OpenCV, and TorchVision (6.82%). In contrast, several packages have installability values of 0.00%, indicating that this category contributes little to separating them in the final ranking.

The quality category is also non-uniform. Most shortlisted packages receive a value of 4.00%, while PyTorch, TensorBoard, and TensorFlow receive 0.00% in this dimension. Overall, the current results indicate that the ranking is driven primarily by a small subset of differentiating categories, while many of the remaining quality categories are comparatively uniform across the shortlisted packages.

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Library	Installability	Quality	Correctness & Verifiability	Surface Reliability	Surface Robustness	Surface Usability	Reusability	Visibility / Transparency	Surface Understandability
Apache MxNet	0.00	4.00	0.00	3.57	3.57	3.57	3.57	3.57	3.57
Caffe	0.00	4.00	0.00	3.57	3.57	3.57	3.57	3.57	3.57
Captum	11.36	4.00	0.00	3.57	3.57	3.57	3.57	3.57	3.57
Catboost	18.18	4.00	0.00	3.57	3.57	3.57	3.57	3.57	3.57
Chainer	0.00	4.00	0.00	3.57	3.57	3.57	3.57	3.57	3.57
CNTK	0.00	4.00	0.00	3.57	3.57	3.57	3.57	3.57	3.57
Cudf	2.27	4.00	0.00	3.57	3.57	3.57	3.57	3.57	3.57
DeepLearning4J	6.82	4.00	0.00	3.57	3.57	3.57	3.57	3.57	3.57
Gensim	4.55	4.00	0.00	3.57	3.57	3.57	3.57	3.57	3.57
Hugging Face Transformers	4.55	4.00	0.00	3.57	3.57	3.57	3.57	3.57	3.57
Jax	4.55	4.00	0.00	3.57	3.57	3.57	3.57	3.57	3.57
Keras	0.00	4.00	0.00	3.57	3.57	3.57	3.57	3.57	3.57
LightGBM	0.00	4.00	0.00	3.57	3.57	3.57	3.57	3.57	3.57
MLFlow	0.00	4.00	0.00	3.57	3.57	3.57	3.57	3.57	3.57
NeuroLab	6.82	4.00	0.00	3.57	3.57	3.57	3.57	3.57	3.57
NLTK	4.55	4.00	0.00	3.57	3.57	3.57	3.57	3.57	3.57
OpenCV	6.82	4.00	0.00	3.57	3.57	3.57	3.57	3.57	3.57
Optuna	2.27	4.00	0.00	3.57	3.57	3.57	3.57	3.57	3.57
PaddlePaddle	0.00	4.00	0.00	3.57	3.57	3.57	3.57	3.57	3.57
Pytensor	2.27	4.00	0.00	3.57	3.57	3.57	3.57	3.57	3.57
PyTorch	0.00	0.00	0.00	3.57	3.57	3.57	3.57	3.57	3.57
Pytorch Geometric	4.55	4.00	0.00	3.57	3.57	3.57	3.57	3.57	3.57
Scikit	2.27	4.00	0.00	3.57	3.57	3.57	3.57	3.57	3.57
SHAP	4.55	4.00	0.00	3.57	3.57	3.57	3.57	3.57	3.57
TensorBoard	4.55	0.00	0.00	3.57	3.57	3.57	3.57	3.57	3.57
TensorFlow	2.27	0.00	99.98	3.57	3.57	3.57	3.57	3.57	3.57
Torch Vision	6.82	4.00	0.00	3.57	3.57	3.57	3.57	3.57	3.57
XGBoost	0.00	4.00	0.00	3.57	3.57	3.57	3.57	3.57	3.57

Figure 3: Heatmap summarizing per-quality scores for the 28 shortlisted neural network software packages. Darker cells indicate larger values within a given category.

4.6. Sensitivity analysis

how rankings change under weight variation, identify stable vs unstable parts of the ranking and summary of implications.

4.7. Popularity vs evidence-based ranking (optional)

Compare evidence-based results against reputation indicators (e.g. stars/download). Discuss alignment/misalignment and what it suggests about state-of-practice versus popularity.

5. Discussion and Recommendations

The current results suggest that the overall ranking is driven primarily by a small subset of differentiating categories, rather than by broad variation across all qualities. In particular, installability and correctness and verifiability contribute strongly to the separation between packages, while several of the remaining surface-level qualities are comparatively uniform across the shortlisted set. This means that, under the current assessment configuration, packages that perform strongly in the most differentiating categories can rise substantially in the overall ranking.

For users of neural network software, the ranking provides a useful evidence-based comparison, but it should not be interpreted as a universal statement that one package is always best for every purpose. Instead, the results are most useful for identifying strong candidates and for showing which quality dimensions currently contribute most to the observed differences across packages. For maintainers, the results highlight areas where visible software-engineering practice can be strengthened and made more consistent.

Several recommendations follow from the current findings. First, maintainers should improve installability by providing clear installation instructions, dependency information, and environment guidance. Second, projects should strengthen correctness and verifiability by exposing more visible evidence of testing, validation, and related quality assurance practices. Third, maintainers should continue to improve transparency through well-maintained public artifacts, since these materials directly affect the evidence available for state-of-practice assessment. Finally, future refinements of the assessment process should consider how to better differentiate categories that are currently highly uniform across the shortlisted packages.

6. Threats to Validity and Limitations

This work is subject to several threats to validity. Construct validity is limited by the fact that the assessment relies on publicly visible artifacts as evidence of software-engineering practice. A package may implement strong internal practices that are not clearly reflected in its repository, documentation, or related public materials. Similarly, categories that appear uniform in the current results may reflect limitations of the available evidence or of the applied scoring scheme rather than true equality across packages.

Internal validity is affected by the need for manual judgment when answering some measurement-template questions. Although the methodology and template provide structure, interpretation of public artifacts can still introduce subjectivity. In addition, the current dataset records package-level references through project website and repository URLs rather than per-question artifact links, which limits the granularity of traceability.

External validity is limited because the assessment focuses on a specific shortlist of 28 open-source neural network software packages. The results therefore should not be generalized automatically to all neural network software or to adjacent communities using general-purpose

supporting libraries. The scope was intentionally restricted to align with the target audience of neural network software developers and maintainers.

Temporal validity is also a limitation. Open-source repositories evolve over time as releases, issues, pull requests, documentation, and tests change. As a result, the assessment reflects the state of the observed artifacts at the time of data collection rather than a permanent ranking. Finally, interviews with developers or maintainers were not conducted at this stage, so the analysis is based only on publicly available evidence and repository-derived measures.

7. Conclusion and Future Work

This paper presented a state-of-practice assessment of 28 open-source neural network software packages using DomainX, a tool-supported workflow based on a published methodology. The work contributes a curated shortlist, a traceable dataset containing measurement-template responses and repository-based measures, and a tool that supports evidence capture, scoring, ranking, and export of assessment artifacts.

The current results show that the overall ranking is shaped primarily by a small subset of differentiating categories, especially installability and correctness and verifiability, while several other quality categories are comparatively uniform across the shortlisted packages. TensorFlow emerged as the top-ranked package under the current configuration, with the remaining packages more tightly clustered below it.

Future work includes refining categories that are currently weakly differentiated, and extending the assessment with additional validation mechanisms such as interviews with developers or maintainers. Further work on DomainX may also improve automation, traceability granularity, and support for repeated assessments as repositories evolve over time.

Appendix A. Assessed packages and exclusions

table listing all assessed packages and excluded candidates with brief reasons to help traceability.

Appendix B. Measurement template excerpt and scoring rules

compact excerpt of measurement template and key scoring/normalization rules, or provide link to full template.

Appendix C. AHP weights and matrices

final AHP weights and any matrices needed for transparency and reproducibility.

Appendix D. DomainX screenshots and exported artifacts

potential figures: ranking view, table view, metric editing view, export example.

References

Spencer Smith, Jacques Carette, Peter Michalski, Ao Dong, and Olu Owojaiye. Methodology for assessing the state of the practice for domain x. *arXiv preprint arXiv:2110.11575v1*, abs/2110.11575:1–15, October 2021. URL `https://arxiv.org/abs/2110.11575`.